

Lab 06

Due March 15, 5:00 PM PST

AUTHOR

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```
library(carData)
data <- data("Prestige")
prestige_data <- Prestige[, c("education", "income")]
```

Modify this template file to create your submission.

Problem 1a

```
## write your answer here
set.seed(100)
prop_train <- 0.75
prop_test <- 1 - prop_train

n_data <- nrow(prestige_data)

train_test <- sample(c("Train", "Test"),
                    size = n_data, replace = TRUE,
                    prob = c(prop_train, prop_test))

train_data <- prestige_data[train_test == "Train", ]
test_data <- prestige_data[train_test == "Test", ]
```

Problem 1b

```
## write answer here

x <- train_data$education
y <- train_data$income
fit <- lm(y ~ x)
fit
```

Call:

```
lm(formula = y ~ x)
```

Coefficients:

(Intercept)	x
-3332.3	973.6

```
est_Y <- predict(fit, newdata = test_data)
```

Warning: 'newdata' had 24 rows but variables found have 78 rows

```
RMSPE <- sqrt( mean( (est_Y - test_data$income) ^2) )
```

Warning in est_Y - test_data\$income: longer object length is not a multiple of shorter object length

```
RMSPE
```

```
[1] 3887.753
```

Problem 1c

```
## write answer here
library(caret)
```

Loading required package: ggplot2

Loading required package: lattice

```
set.seed(100)
K_vals <- seq(from = 1, to = 50, by = 1)
K_RMSPE <- rep(NA, length(K_vals))

CV_folds <- sample(1:5, size = nrow(train_data), replace = TRUE)
num_folds <- 5

for(j in 1:length(K_vals)) {
  k <- K_vals[j]
  CV_RMSPE <- rep(NA, num_folds)

  for(i in 1:num_folds){
    cv_train <- train_data[CV_folds != i, ]
    cv_val <- train_data[CV_folds == i, ]

    fit_knn <- knnreg(income ~ education, data = cv_train, k = k)
    cv_pred <- predict(fit_knn, cv_val)

    CV_RMSPE[i] <- sqrt(mean(((cv_pred - cv_val$income))^2))
  }

  K_RMSPE[j] <- mean(CV_RMSPE)
}
```

```
best_k <- K_vals[which.min(K_RMSPE)]  
best_k
```

```
[1] 10
```

Problem 1d

```
## write your answer here
```

```
fit_knn <- knnreg(income ~ education, data = train_data, k = best_k)  
pred <- predict(fit_knn, test_data)  
sqrt(mean((pred - test_data$income)^2))
```

```
[1] 2189.479
```